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**Week 2: Image Formation**

- **Reading:** Week 2 Lecture Handout — 30 min
- **Ungraded Plugin:** 2.1 Overview of Image Formation — 3 min
- **Practice Quiz:** 2.1 Overview of Image Formation Self-Check Quiz — Started
- **Ungraded Plugin:** 2.2 Pinhole and Perspective Projection — 21 min
- **Practice Quiz:** 2.2 Pinhole and Perspective Projection Self-check Quiz — Started
- **Reading:** 2.3 Video Correction — 10 min
- **Ungraded Plugin:** 2.3 Image Formation using Lenses — 12 min
- **Practice Quiz:** 2.3 Image Formation using Lenses Self-check Quiz — Started
- **Ungraded Plugin:** 2.4 Depth of Field — 15 min
- **Practice Quiz:** 2.4 Depth of Field Self-check Quiz — 15 min
- **Ungraded Plugin:** 2.5 Lens Related Issues — 8 min
- **Practice Quiz:** 2.5 Lens Related Issues Self-check Quiz — 5 min
- **Ungraded Plugin:** 2.6 Wide Angle Cameras — 24 min
- **Ungraded Plugin:** 2.7 Animal Eyes — 16 min
- **Discussion Prompt:** Week 2 Intrusive Detection System — 10 min

**Week 2 Quiz**

## Congratulations! You passed!

Go to next item

**Grade received** 100% **To pass** 50% or higher

# 2.3 Image Formation using Lenses Self-check Quiz

**1.** An image can only be formed on the focal point. **1 / 1 point**

- ○ True
- ◉ False

coach

Ready to review what you've learned before taking the quiz? I'm here to help.

Help me practice Let's chat

**Correct**
An image is formed on the focal point only if the object is infinite far away and on the optical axis. If it is infinitely far away but not on the optical axis, then the image is formed on the focal plane, a plane on the focal point and perpendicular to the optical axis.
If the object is not infinitely far away, then the image is formed at a distance from the lens given by the formula: $1/F = 1/A + 1/B$, where $F$ is the focal distance, $A$ the distance object-lens and $B$ the distance image-lens.

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To Pass 50% or higher

Your grade 100%

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We keep your highest score

**2.** Lenses perform perspective projection. **1 / 1 point**

- ◉ True
- ○ False

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**Correct**
All the properties of the perspective projection still apply to lenses.

(Refer to this link to learn more about perspective projection)

**3.** Why do we use lenses when we have pin-hole camera? **1 / 1 point**

- ○ Cameras with lenses are easier to manufacturer when compared to pin-hole cameras
- ◉ Lenses gather more light than pin-hole cameras, reducing exposure time
- ○ Lenses can be stacked to produce a clear image of the scene
- ○ Lenses keep parallel lines from the scene, parallel in the image

**Correct**
Lenses have a wide surface area to collect more light and the geometry of the lens allows multiple light rays to converge to the same point in the 2D plane. This allows cameras with lenses to collect sufficient light to reduce exposure time to a fraction of a second.

**4.** How many planes of focus does a fixed lens have? **1 / 1 point**

- ◉ 1
- ○ This depends on the object's distance from the lens in the scene
- ○ There are multiple; however, an exact number cannot be defined
- ○ Infinite

**Correct**
For a given fixed lens, there is only one plane of focus. If we move the lens the plane of focus changes; however, there can only be one plane of focus.